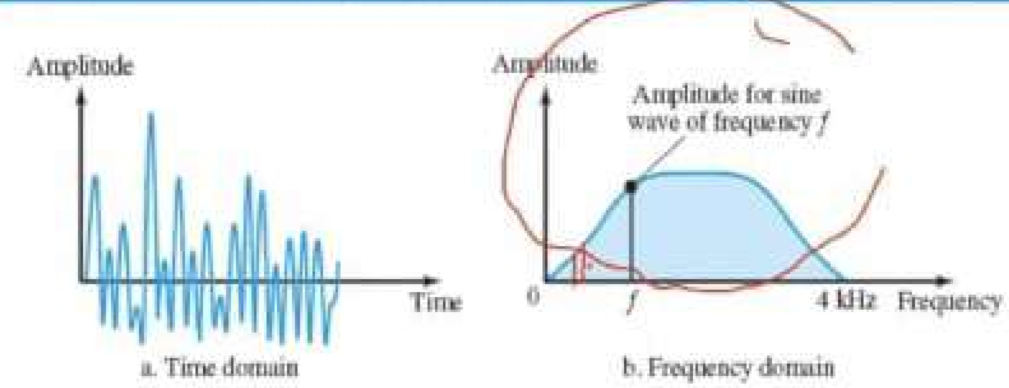
1. In data communications we often use the following frequency components:

a) f,2f,3f,... b) 1.2f,1.3f,4.5f ,... c) f,3f.5f,7f,...

A single-frequency sine wave is not useful in data communications; we need to send a composite signal, a signal made of many simple sine waves.

Figure 3.12 *The time and frequency domains of a nonperiodic signal*



2. To make a code with no dc component, each bit must has:

a) positive & negative voltages

b) One kind of voltages

c) Any kind of voltages

3. ISP refers to:

a) Internet Subscriber Protocol
b) Internet Service Protocol
c) Internet Service Provider
d) Internet Service Process

Backbones and provider networks are also called **Internet Service Providers (ISPs).** The backbones are often referred to as *international ISPs;* the provider networks are often referred to as *national* or *regional ISPs*.

4. 0&1 refer in data communications as:

a) digital data
b) digital signals
c) decimal numbers
d) voltage levels

Digital data take on discrete values. For example, data are stored in computer memory in the form of 0s and 1s. They can be converted to a digital signal or modulated into an analog signal for transmission across a medium.

5. $y=3\sin(2x-\pi)$, has phase shift equal to:

a) $-\pi/2$ b) $-\pi$ c) $\pi/2$ d) π

$$2x-\pi=0$$

$$x=\frac{\pi}{2}$$

6. If we want to allocate TV stations (analog black and white) into FM band, then we can put:

a) not more than 3 stations　　　b) less than 3 stations

c) more than 3 stations　　　d) cannot be calculated

Example 3.15

The bandwidth needed is 5.5124 MHz. $\approx 6\,MHz$

$$88 - 108\ MHz \longrightarrow \Delta f = \frac{20}{6}\ MHz \simeq 3{,}3 \approx \underline{3}$$

7. The concepts of baseband transmission & broadband transmission are related to:
   a) digital signals only
   b) analog signals only
   c) digital& analog signals
   d) composite signals only

8. How much power in -26 dBm?

a) -2.5mw
b) -2.5 µw
c) 2.5mw
d) 2.5 µw

$$P(dBm) = -26 = 10 \log \frac{P}{1mW}$$

$$-2,6 = \log \frac{P}{1mW}$$

$$10^{-2,6} = \frac{P}{1mW}$$

$$2,5 \times 10^{-3} = \frac{P}{1mW}$$

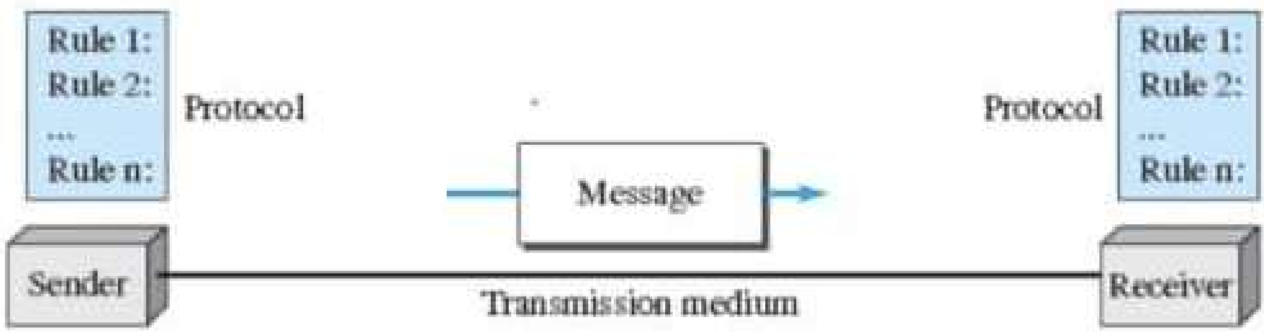
$$2,5 \times 10^{-3} \times 1 \times 10^{-3} = P$$

$$2,5 \times 10^{-6}\,W = P$$

$$2,5\ \mu W = P$$

9. The following is not component in data communication system:
a) video clip b) TCP/IP c) air d) broadcasting

1.1.1 Components



10. The three primary colors are:
a) RGB b) RJB c) RYG d) BMY

There are several methods to represent color images. One method is called **RGB**, so called because each color is made of a combination of three primary colors: *r*ed, *g*reen, and *b*lue. The intensity of each color is measured, and a bit pattern is assigned to

11. The physical layer of a network:
   a) constructs packets of data & sends them across the network
   b) controls error detection & correction
   c) defines the electrical characteristics of signals passed between the computer &communication device
   d) all the above

12. In network terminology, the end system is called:

a) host b) node c) modem d) switch

13. The following is connecting device:

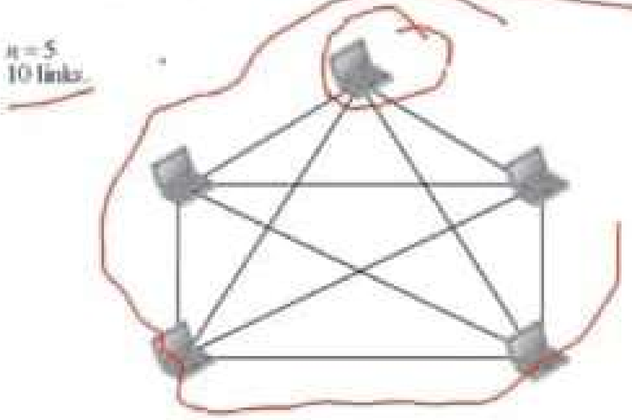
a) host b) router c) sender d) receiver

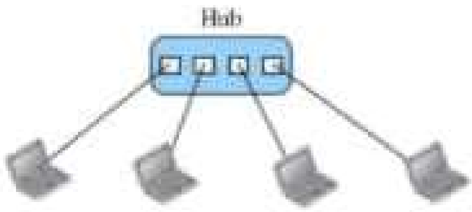
14. What does the acronym ISDN stands for:

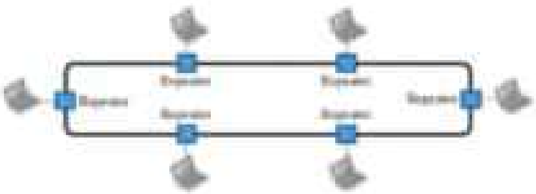
a) Intelligent Standard Digital Network
b) Intelligent Services Digital Network
c) Integrated Services Digital Network
d) Integrated Services Data Network

15. In this topology, all links are dedicated:

a) ring b) mesh c) bus d) hybrid







16. In 8B6T code, there is:

a) 473 extra signal elements for synchronization & error detection

b) 573 extra signal elements for synchronization & error detection

c) 24 extra signal elements for synchronization & error detection

d) 2 extra signal elements for synchronization & error detection

8B6T

A very interesting scheme is **eight binary, six ternary (8B6T).** This code is used with 100BASE-4T cable, as we will see in Chapter 13. The idea is to encode a pattern of 8 bits as a pattern of six signal elements, where the signal has three levels (ternary). In this type of scheme, we can have $2^8 = 256$ different data patterns and $3^6 = 729$ different signal patterns. The mapping table is shown in Appendix F. There are $729 - 256 = 473$ redundant signal elements that provide synchronization and error detection. Part of the

Q2: Consider an audio signal with spectral components in range (300 to 3000) Hz. Assume that a sampling rate of 7000 samples per second will be used to generate a PCM signal:

a) For SNR=30 dB, what is the number of uniform quantization levels needed?
b) What data rate is required?
c) If a signal of expression: $x(t)=3\cos 50\pi t + 10\sin 300\pi t - \cos 100\pi t$
Calculate Nyquist rate for this expression.

Q2 :

$$\boxed{SNR_{dB} = 6.02\, n_b + 1.76 \text{ dB}}$$

a)

$$n_b = \frac{SNR - 1.76}{6.02} = \frac{30 - 1.76}{6.02} = 4.69 \approx 5 \text{ bits} \rightarrow +2$$

$$\text{quantization levels} = 2^{n_b} = 2^5 = 32 \text{ quantization levels} \rightarrow +2$$

b)

$$N = \text{number of Samples/s} \times \text{bits/Sample}$$
$$= 7000 \times 5 = 35 \text{ Kbps} \rightarrow +1$$

c)

$$\omega_2 t = 300\pi t \rightarrow f_{max} = 150 Hz$$
$$f_{sampling} = 2 f_{max} = 300 Hz$$

+1